

Cooper Le-Doux

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Education

- **M.S. of Biomedical Engineering** (In progress, May 2027) – Univ. of Colorado Boulder
- **B.S. of Chemical Engineering** (Aug. 2020 – May 2025), GPA: 3.53 – New Mexico State Univ.
 - Minors: Biomedical Eng., Computational Eng., Electrical Eng., Chemistry

Skills

- **Technical:** Python | MATLAB | C | R | EEG Analysis | Microscopy | Cell Culture | MNE-Python

Key Experience

- **Lawrence Livermore National Laboratory – Research Intern**

National Nuclear Security Administration Minority Serving Institutions Internship Program

- **EEG CNN Research (Aug 2023 – Aug 2025)**

- Improved Python (matplotlib, NumPy, PyTorch) while strengthening problem-solving and persistence
- Explored machine learning algorithms and OpenNeuro/MNE-Python datasets, showing adaptability and independent learning
- Advanced machine learning knowledge toward decision tree modeling for VR simulation, reflecting long-term research commitment

- **Cryptographic Analysis (Aug 2022 – Apr 2023)**

- Trained in Bash, R, and C for Intel/AMD architectures, demonstrating versatility and technical adaptability
- Analyzed side-channel attacks on DOE supercomputers and contributed to a new latency API, applying critical thinking, initiative, and teamwork

- **Hydrogel Analysis (Jun 2022 – Aug 2022)**

- Performed microbead labeling, imaging, and cell culture, showcasing precision and reliability in lab work
- Automated microbead processing in MATLAB (90% faster), highlighting creativity and efficiency, and communicated results in meetings/competitions

- **Native Camp – Online English Instructor**

- Taught English to international learners, adapting to diverse ages, accents, and technical challenges. Strengthened patience, communication, and cultural awareness.

- **New Mexico Ranch & Heritage Museum – Blacksmith Volunteer**

- Engaged wide audiences in demonstrations while ensuring safety and accessibility of technical knowledge.

Additional

Languages: English (native), Spanish (secondary)

Certifications: Teaching English as a Foreign Language (TEFL)